

POSSESSION OF MOBILES IN EXAM IS UFM PRACTICE.

Name Riya Verma

Enrollment No. 23803012

Jaypee Institute of Information Technology, Noida
End Semester Examination, 2024 (Even)
B.Tech, II Semester

Course Title: Software Development Fundamentals-II
Course Title: 15B11CI211

Maximum Time: 02 Hr
Maximum Marks: 35

Course Outcomes	
CO1	Explain various object-oriented concepts like class and objects, constructor, abstraction, data hiding (access specifiers etc.)
CO2	Apply object-oriented concepts like inheritance, polymorphism and templates to real-world problems.
CO3	Apply SQL commands to create tables and perform various operations like insert, delete, select on the tables.
CO4	Analyze the source code for possible outcomes, exceptions and to debug the errors.
CO5	Design and implement class diagram for varied real life problems using OOPs.

Note: Attempt all questions.

Q1: Suppose you are organising the 'car of the year' poll and hence you want to store the rating (1-5) for each car along with the car name. A car can have any number of rating associated with it. However, the overall rating of the car is the average of all the rating it has. Write a program in C++ that can serve the following functionality using an appropriate STL container:

- Store the rating of at least three cars.
- Display the car with highest average rating as car of the year.

[CO2(Applying), 6 Marks]

Q2: Write appropriate SQL statements to perform the following tasks:

PaintingName	Artist	Year	Country	Valuation
Monalisa	Leonardo da vinci	1506	Italy	860
Water Lilies	Claude Monet	1920	France	600
The last Judgement	Michelangelo	1741	Italy	700
Interior	Edger Degas	1869	France	543

- Create the given relation 'Painting' where primary key is a combination of Artist, Year & Country. Populate the table with any two records provided in the table given above.
- Add a new column named 'Period' to the above table. Set the Period of all the paintings belonging to the year 1500 to 1600 as "Renaissance" and remaining ones with the value "Impressionism".
- Display the average valuation of paintings from each country.
- Display all the Italian paintings whose valuation is above 500.
- Display all the artists whose has string "on" in their names and belongs to either Italy or France.

[CO3 (Applying), 2+2+1+1+1=7 Marks]

Q3: Create a generic function with (arg1, arg2) that is capable of returning the sum of two same type of arguments. Demonstrate the generic nature of this sum function by calling it with two integer arguments and two complex number arguments where complex is a user defined generic class with two data members (real and imaginary) each of same type. You can use additional functions if needed.

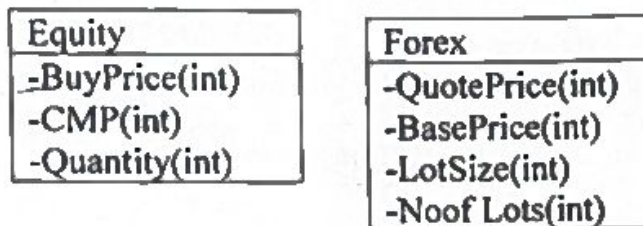
[CO2(Applying), 5 Marks]

Q4: Create a class called Fraction having two data members: num and dino [both are integers]. Write a C++ program that can input n fractions from the user, store them in a dynamically

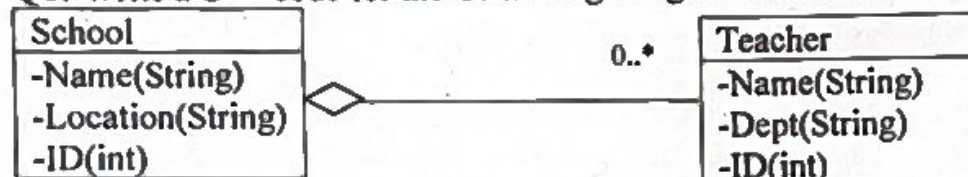
created array and display their product. Analyse the user input, in case the user enters zero for the denominator of any fraction object, the program will terminate by throwing an appropriate exception.
[CO4 (Analysing), 5 Marks]

Q5: Equity and forex are two popular financial instruments where investors can invest their money. Create the following classes and design a function in C++ that can take an instance of each of these class and display the details of the better yielding financial instrument. The yield of an equity is $((BuyPrice - CMP) * Quantity) / BuyPrice$ while the yield of the forex is $((QuotePrice - BasePrice) * LotSize * Noof Lots) / (QuotePrice + BasePrice)$. You can add additional getter, setter and other methods as per your convenience.

[CO1 (Understanding), 4 Marks]



Q6: Write a C++ code for the UML diagram given below & include the suitable Methods.



[CO5 (Creation), 4 Marks]

Q7: Analyse the given code & find its output. Give suitable interpretation of the results obtained. Assume there is no syntax error.
[CO4 (Analysing), 4 Marks]

```

a)
#include<iostream>
using namespace std;
class A{
public:
    int x;
    A(){cout<<"A Default"<<endl;}
    A(int i){x=i; cout<<"A para"<<endl;};
    class B:virtual public A{
    public:
        B(): A(7){cout<<"B default"<<endl;}
        B(int j):A(4) { cout<<"B para"<<endl;}
        void print() { cout<<x;};

    class D:public B{
    public:
        D():B(3) { cout<<"D default"<<endl;};

    int main(){
        D d;
        d.print();
        return 0;}
    
```

```

b)
#include<iostream>
#include<fstream>
#include<string>
#include<cctype>
using namespace std;
int main()
{
    char ch='a';
    ifstream file("MyPoem.txt")
    int mcount=0, lcount, i=0;
    string line;
    while(getline(file, line)){
        int count=0;
        for(char c:line){
            if(tolower(c) == tolower(ch)){
                count++;
            }
            i++;
        }
        if (count > mcount){
            mcount = count;
            lcount = i;} }
    file.close();
    cout<<mcount<<" present in "<<lcount;
    return 0;}
    
```

Content of file "MyPoem.txt" :
 Shall I compare thee summer day?
 And Summer's lease hath all too short a date
 Thou art lovelier and more temperate
 Rough winds do shake the darling buds of May